Indore Institute of Science & Technology

Affiliated to - RGPV (Bhopal) & Approved by - AICTE (New Delhi)

Program: Bachelor of Technology (B.Tech.)

Branch: AI & ML

Subject Name: Cloud Computing
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Syllabus

UNIT I Introduction of Grid and Cloud computing, characteristics, components, business and IT perspective, cloud services requirements, cloud models, Security in public model, public verses private clouds, Cloud computing platforms: Amazon EC2,Platform as Service: Google App Engine, Microsoft Azure, Utility Computing, Elastic Computing.

UNIT II Cloud services- SAAS, PAAS, IAAS, cloud design and implementation using SOA, conceptual cloud model, cloud stack, computing on demand, Information life cycle management, cloud analytics, information security, virtual desktop infrastructure, storage cloud.

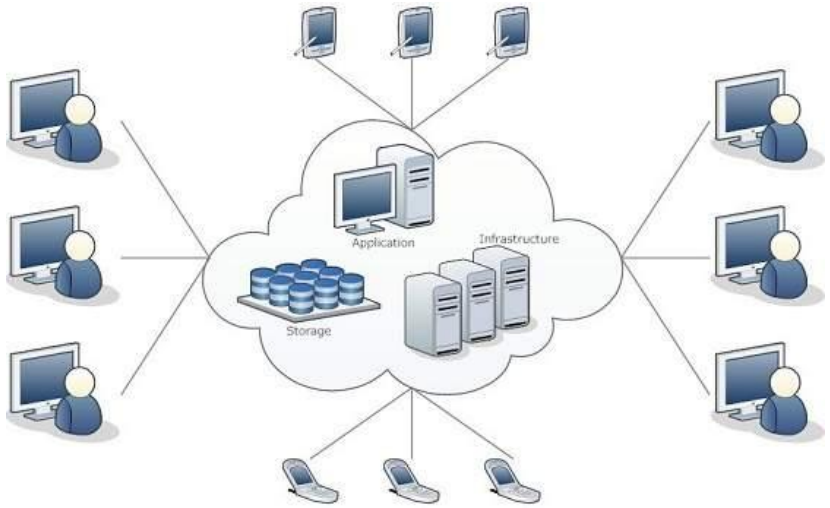
UNIT III Virtualization technology: Definition, benefits, sensor virtualization, HVM, study of hypervisor, logical partitioning- LPAR, Storage virtualization, SAN, NAS, cloud server virtualization, virtualized data center.

UNIT IV Cloud security fundamentals, Vulnerability assessment tool for cloud, Privacy and Security in cloud, Cloud computing security architecture: Architectural Considerations- General Issues, Trusted Cloud computing, Secure Execution Environments and Communications, Micro-architectures; Identity Management and Access control-Identity management, Access control, Autonomic Security, Cloud computing security challenges: Virtualization security management-virtual threats, VM Security Recommendations, VM-Specific Security techniques, Secure Execution Environments and Communications in cloud.

UNIT V SOA and cloud, SOA and IAAS, cloud infrastructure benchmarks, OLAP, business intelligence, e-Business, ISV, Cloud performance monitoring commands, issues in cloud computing. QOS issues in cloud, mobile cloud computing, Inter cloud issues, Sky computing, Cloud Computing Platform, Xen Cloud Platform, Eucalyptus, OpenNebula, Nimbus, TPlatform, Apache Virtual Computing Lab (VCL), Anomaly Elastic Computing Platform.

Introduction to Cloud Computing

- **Cloud Computing** is the delivery of computing services such as servers, storage, databases, networking, software, analytics, intelligence, and more, over the Cloud (Internet).
- Cloud Computing refers to **manipulating, configuring,** and **accessing** the hardware and software resources remotely. It offers online data storage, infrastructure, and application.
- Cloud computing offers **platform independency,** as the software is not required to be installed locally on the PC. Hence, the Cloud Computing is making our business applications **mobile** and **collaborative.**



Definition

- **NIST's Definition of Cloud Computing** - *"A model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort or service provider interaction. This cloud model is composed of five essential characteristics, three service models, and four deployment models."*

- **IBM's Definition of Cloud Computing** – "Cloud Computing often referred to as simply "the cloud" is the delivery of on-demand computing resources – everything from applications to data centers- over the internet on a pay-for-use basis.

Characteristics of Cloud Computing

1. **On-demand self-services:**
   Users can request for resources whenever the need arises and can manage the allocated resources on their own without much interaction from service providers.

2. **Broad network access:**
   The Computing services are generally provided over standard networks and can be accessed using an device such as smart phones, laptops, etc.

3. **Rapid elasticity:**
   The Computing services should have IT resources that are able to scale out and in quickly and on as needed basis. Whenever the user require services it is provided to him and it is scale out as soon as its requirement gets over.

Characteristics of Cloud Computing

4. **Resource pooling:**
   The IT resource (e.g., networks, servers, storage, applications, and services) present are shared across multiple applications and occupant in an uncommitted manner. Multiple clients are provided service from a same physical resource.

5. **Measured service:**
   The resource utilization is tracked for each application and occupant, it will provide both the user and the resource provider with an account of what has been used. This is done for various reasons like monitoring, billing and effective use of resource.

Architecture of Cloud Computing

- Cloud Computing Architecture is a combination of components required for a Cloud Computing service. A Cloud computing architecture consists of several components which are classified under frontend platform and a backend platform.

- The front end consists of the client part of a cloud computing system. It comprises interfaces and applications that are required to access the Cloud computing or Cloud programming platform.

- The back end refers to the cloud itself, it comprises the resources required for cloud computing services. It consists of virtual machines, servers, data storage, security mechanisms, etc.

Architecture of Cloud Computing

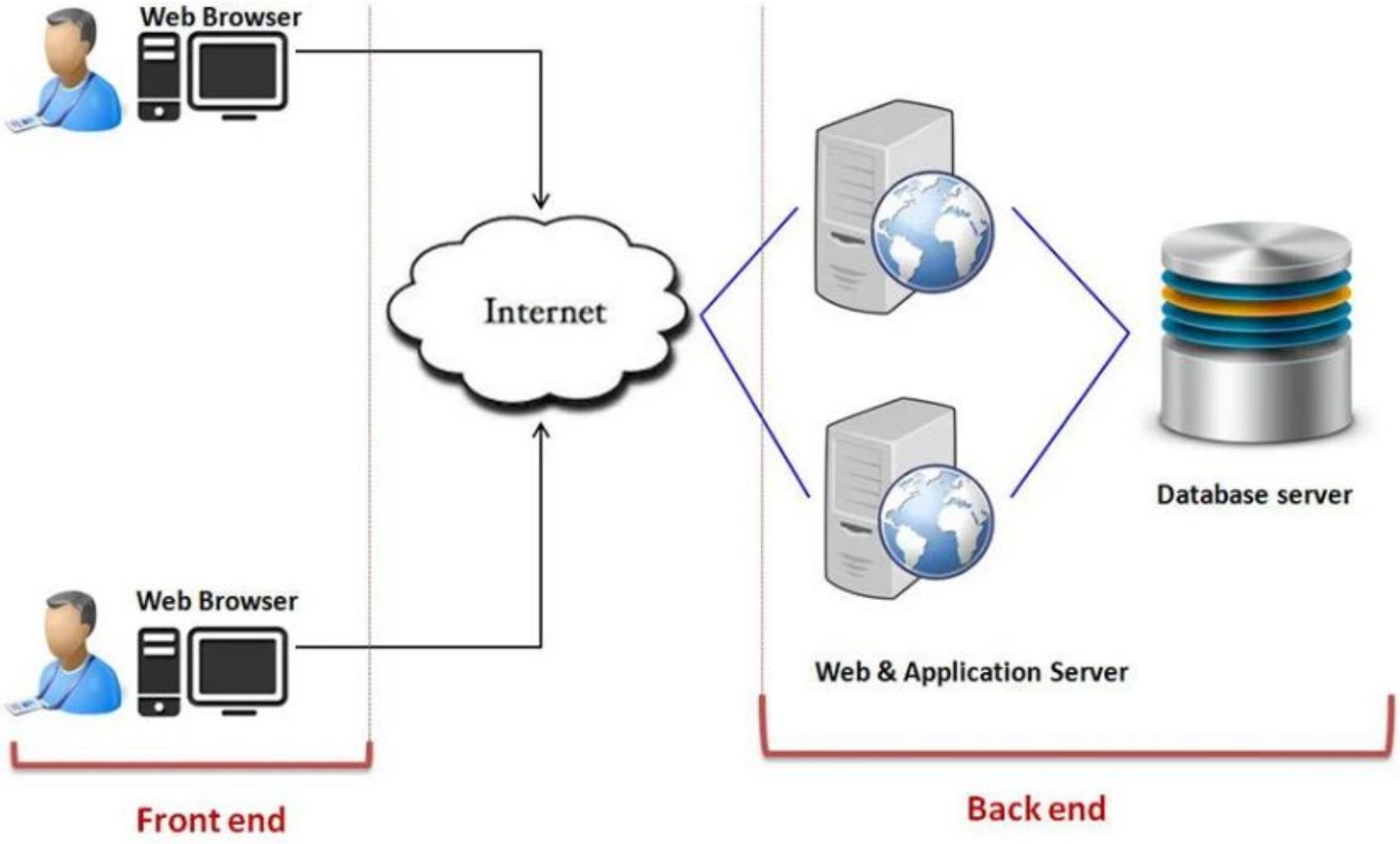


Fig. Architecture of Cloud Computing

Components of Cloud Computing

1. Client Infrastructure:
Client Infrastructure is a front-end component that provides a GUI. It helps users to interact with the Cloud.

2. Application:
The application can be any software or platform which a client wants to access.

3. Service:
The service component manages which type of service you can access according to the client's requirements.

Components of Cloud Computing

Three Cloud computing services are:

- Software as a Service (SaaS)
- Platform as a Service (PaaS)
- Infrastructure as a Service (IaaS)

4. Runtime Cloud:

Runtime cloud offers the execution and runtime environment to the virtual machines.

5. Storage:

Storage is another important Cloud computing architecture component. It provides a large amount of storage capacity in the Cloud to store and manage data.

Components of Cloud Computing

6. Infrastructure:

It offers services on the host level, network level, and application level. Cloud infrastructure includes hardware and software components like servers, storage, network devices, virtualization software, and various other storage resources that are needed to support the cloud computing model.

7. Management:

This component manages components like application, service, runtime cloud, storage, infrastructure, and other security matters in the backend. It also establishes coordination between them.

Components of Cloud Computing

8. Security:

Security in the backend refers to implementing different security mechanisms for secure Cloud systems, resources, files, and infrastructure to the end-user.

9. Internet:

Internet connection acts as the bridge or medium between frontend and backend. It allows you to establish the interaction and communication between the frontend and backend.

Cloud Service Requirements

- **Heterogeneous Systems Support**
- **Service Management**
- **Dynamic Workload and Resource Management**
- **Reliability, Availability & Security**
- **Integration with Data Center Management Tools**
- **Visibility & Reporting**
- **Administrator, Developer and End User Interfaces**

Cloud Service Requirements

- **Service and Resource Management :** A cloud infrastructure virtualizes all components of a data center. Service management is a measured package of applications and services that end users can easily deploy and manage via a public and/or private cloud vendor. And a simplified tool to outline and gauge services is vital for cloud administrators to market functionality. Service management needs to contain resource maintenance, resource guarantees, billing cycles, and measured regulations. Once deployed, management services should help create policies for data and workflows to make sure it's fully efficient and processes are delivered to systems in the cloud.

- **Data Center Management Tools Integration:** Most data centers utilize a variety of IT tools for systems management, security, provisioning, customer care, billing, and directories, among others. And these work with cloud management services and open APIs to integrate existing operation, administration, maintenance, and provisioning (OAM&P) systems. A modern cloud service should support a data center's existing infrastructure as well as leveraging modern software, hardware, and virtualization, and other technology.

Cloud Service Requirements

- **Reporting, Visibility, Reliability & Security:** Data centers need high levels of real-time reporting and visibility capabilities in cloud environments to guarantee compliance, SLAs, security, billing, and chargebacks. Without robust reporting and visibility, managing system performance, customer service, and other processes are nearly impossible. And to be wholly reliable, cloud infrastructures must operate regardless of one or more failing components. For to safeguard the cloud, services must ensure data and apps are secure while providing access to those who are authorized.

- **Interfaces for Users, Admins, and Developers:** Automated deployment and self-service interfaces ease complex cloud services for end users, helping lower operating costs and deliver adoption. Self-service interfaces offer customers the ability to effectively launch a cloud service by managing their own data centers virtually, designing and driving templates, maintaining virtual storage, networking resources, and utilizing libraries. Administrator interfaces present better visibility to all resources, virtual machines, templates, service offers, and various cloud users. And all of these structures integrate by way of APIs for developers.

Types of Cloud Computing

There are certain services and models working behind the scene making the cloud computing feasible and accessible to end users. Following are the working models for cloud computing:

- **Deployment Models**
- **Service Models**

Deployment Models

Deployment models define the type of access to the cloud, i.e., how the cloud is located? Cloud can have any of the three types of access: **Public, Private, Hybrid, (Community).**

Service Models

Cloud computing is based on service models. These are categorized into three basic service models which are - (IaaS), (PaaS), (SaaS)

Types of Cloud Computing(Deployment Models)

- **Public cloud:** Public clouds are owned and operated by a third-party cloud service providers, which deliver their computing resources, like servers and storage, over the Internet. Microsoft Azure is an example of a public cloud. With a public cloud, all hardware, software, and other supporting infrastructure is owned and managed by the cloud provider. You access these services and manage your account using a web browser.

- **Private cloud:** A private cloud refers to cloud computing resources used exclusively by a single business or organization. A private cloud can be physically located on the company's on-site datacentre. Some companies also pay third-party service providers to host their private cloud. A private cloud is one in which the services and infrastructure are maintained on a private network.

- **Hybrid cloud:** Hybrid clouds combine public and private clouds, bound together by technology that allows data and applications to be shared between them. By allowing data and applications to move between private and public clouds, a hybrid cloud gives your business greater flexibility, more deployment options, and helps optimize your existing infrastructure, security, and compliance.

Public versus Private Cloud

	Public Cloud Computing	Private Cloud Computing
Infrastructure	Multiple users – shared network managed by a service provider hosted off-site	Single user – dedicated network managed by the in-house technical team
Requirements	Off-premises	On-premises/off-premises
Scalability	Easily scalable as per the requirement	Depends on the service provider agreement
Maintenance	By the cloud-service provider	By the in-house technical team
Expense	The most affordable cloud that also offers a pay as you go service	High investment cost to set up the network and staff with regular maintenance
Performance	Multiple users might reduce the performance level	High performance for a dedicated server
Uses	Online office applications, storage, and testing of development	High computing power, data protection, and security
Example	Amazon Web Services (AWS)	VMWare, Red Hat

Advantages of Cloud Computing

- **Cost:** It reduces the huge capital costs of buying hardware and software.
- **Speed:** Resources can be accessed in minutes, typically within a few clicks.
- **Scalability:** We can increase or decrease the requirement of resources according to the business requirements.
- **Productivity:** While using cloud computing, we put less operational effort. We do not need to apply patching, as well as no need to maintain hardware and software. So, in this way, the IT team can be more productive and focus on achieving business goals.
- **Reliability:** Backup and recovery of data are less expensive and very fast for business continuity.
- **Security:** Many cloud vendors offer a broad set of policies, technologies, and controls that strengthen our data security.

Disadvantages of Cloud Computing

- **Internet Connectivity:** In cloud computing, every data (image, audio, video, etc.) is stored on the cloud, and we access these data through the cloud by using the internet connection. If you do not have good internet connectivity, you cannot access these data. However, we have no any other way to access data from the cloud.
- **Vendor lock-in:** Vendor lock-in is the biggest disadvantage of cloud computing. Organizations may face problems when transferring their services from one vendor to another. As different vendors provide different platforms, that can cause difficulty moving from one cloud to another.
- **Limited Control:** Cloud infrastructure is completely owned, managed, and monitored by the service provider, so the cloud users have less control over the function and execution of services within a cloud infrastructure.
- **Security:** Although cloud service providers implement the best security standards to store important information. But, before adopting cloud technology, you should be aware that you will be sending all your organization's sensitive information to a third party, i.e., a cloud computing service provider. While sending the data on the cloud, there may be a chance that your organization's information is hacked by Hackers.

Utility Computing

- **Utility computing** is the process of providing computing service through an on-demand, pay-per-use billing method. Utility computing is a computing business model in which the provider owns, operates and manages the computing infrastructure and resources, and the subscribers accesses it as and when required on a rental or metered basis.
- Utility computing is one of the most popular IT service models, primarily because of the flexibility and economy it provides. This model is based on that used by conventional utilities such as telephone services, electricity and gas.
- The principle behind utility computing is simple. The consumer has access to a virtually unlimited supply of computing solutions over the Internet or a virtual private network, which can be sourced and used whenever it's required. The back-end infrastructure and computing resources management and delivery is governed by the provider.
- Utility computing solutions can include virtual servers, virtual storage, virtual software, backup and most IT solutions.
- Cloud computing, grid computing and managed IT services are based on the concept of utility computing.

Elastic Computing

- **Elastic computing** is a concept in cloud computing in which computing resources can be scaled up and down easily by the cloud service provider. Elastic computing is the ability of a cloud service provider to provision flexible computing power when and wherever required. The elasticity of these resources can be in terms of processing power, storage, bandwidth, etc.
- Cloud computing is about provisioning on-demand computing resources with the simplicity of a mouse click.
- Besides providing these services on demand basis, the resources are elastic in nature, i.e. they can be easily scaled depending upon the underlying resource requirements on run time without even disrupting the operations and this ability is known as elastic computing.
- On a small scale this is done manually, but for larger installations, the scaling is automatic. For example, a larger provider of online video could setup a system so that the number of webservers online scaled during peak viewing hours.

Grid Computing

- **Grid computing** is a distributed system that connects geographically dispersed, heterogeneous computers via networks to act as a virtual supercomputer. By pooling resources like processing power and storage, it enables the collaborative handling of massive, complex tasks—such as scientific simulations or financial modeling—that would be difficult for a single machine.

How It Works

- A user submits a task to a control node, which breaks the job into smaller, independent subtasks. These subtasks are distributed among available provider nodes (computers) in the grid. Each node processes its portion of the work and returns the results to the control node, which then combines them into the final output.

Key Characteristics and Components of Grid Computing

- **Decentralized Nodes:** Computers (nodes) in the grid can be located anywhere, often across different organizations or locations.
- **Virtual Supercomputer:** Multiple computers work together to accomplish a single, shared goal
- **Middleware:** Specialized software that manages resources, enables communication between different nodes, and schedules jobs.
- **Heterogeneous:** The computers in a grid often have different operating systems and hardware configurations.
- **Resource Sharing:** It primarily focuses on sharing computer power, data, and software rather than just files.

THANK YOU!